

Zeta Function

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1 Definition

Definition 1.1. *The Riemann Zeta function is defined as*

$$\zeta(z) = \sum_{k=1}^{\infty} \frac{1}{k^z}$$

We have already shown that ¹

$$\zeta(2) = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \quad (1)$$

2 Bernoulli Numbers

Examining the Taylor expansion of $\frac{x}{e^x-1}$, this will be required for the evaluation of the even integer values of ζ .

Definition 2.1. *The Bernoulli numbers B_n are defined as the coefficients of the Taylor expansion of the following function*

$$\boxed{\frac{x}{e^x-1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n}$$

Theorem 2.1.

$$\sum_k^n \binom{n+1}{k} B_k = 0 \quad \forall n > 0, \text{ with } B_0 = 1 \quad (2)$$

Proof

Starting from definition ??

$$\frac{x}{e^x-1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

¹See Gamma function document

Using the Taylor expansion of e^x

$$\frac{x}{\sum_{k=1}^{\infty} \frac{x^k}{k!}} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$\begin{aligned} x &= \sum_{k=1}^{\infty} \frac{x^k}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n \\ 1 &= \sum_{k=1}^{\infty} \frac{x^{k-1}}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x \end{aligned}$$

Adjusting the indicies

$$1 = \sum_{k=0}^{\infty} \frac{x^k}{(k+1)!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Using Cauchy's product formula for infinite sums ²

$$1 = \sum_{n=0}^{\infty} \underbrace{\sum_{k=0}^n \underbrace{\left(\frac{B_k}{k!} x^k\right)}_{a_k} \underbrace{\left(\frac{x^{n-k}}{(n-k+1)!}\right)}_{b_{n-k}}}_{c_n}$$

Simplifying

$$\begin{aligned} 1 &= \sum_{n=0}^{\infty} \frac{1}{(n+1)!} \sum_{k=0}^n \binom{n+1}{k} B_k x^k x^{n-k} \\ 1 &= \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!} \sum_{k=0}^n \binom{n+1}{k} B_k \end{aligned}$$

Comparing coefficients of powers of x of both sides we get

$$\begin{aligned} n = 0, \quad 1 &= B_0 \\ n \neq 0, \quad 0 &= \sum_{k=0}^n \binom{n+1}{k} B_k \end{aligned}$$

2.1 First few Bernoulli numbers

Using equation ?? with $n = 1$ implies $\binom{2}{0} B_0 + \binom{2}{1} B_1 = 0 = 1 + 2B_1$ which implies

$$B_1 = -\frac{1}{2} \tag{3}$$

² $\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n$ where $c_n = \sum_{k=0}^n a_k b_{n-k}$. See Mertens' Theorem (proof in Bernoulli Numbers document).

Using equation ?? with $n = 2$ implies $\binom{3}{0}B_0 + \binom{3}{1}B_1 + \binom{3}{2}B_2 = 0 = 1 - \frac{3}{2} + 3B_2$ which implies

$$B_2 = \frac{1}{6} \quad (4)$$

Similarly when we apply this equation for increasing values of n

$$B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0, B_6 = \frac{1}{42}, \dots \quad (5)$$

Theorem 2.2. In fact we can show that all odd Bernoulli numbers after $n=1$ are 0

$$B_n = 0 \quad \forall \text{ odd } n > 1 \quad (6)$$

Proof

Starting from definition ??

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Removing the first two terms of the sum

$$\frac{x}{e^x - 1} = B_0 + B_1 x + \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$\begin{aligned} \frac{x}{e^x - 1} - 1 + \frac{x}{2} &= \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n \\ \frac{x + xe^x}{2e^x - 2} - 1 &= \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n \end{aligned}$$

If we can show that the RHS is an even function then we have established the proof

$$\begin{aligned} y(x) &= \frac{x + xe^x}{2e^x - 2} - 1 \\ y(-x) &= \frac{-x + -xe^{-x}}{2e^{-x} - 2} - 1 \end{aligned}$$

Simplifying

$$\begin{aligned} y(-x) &= \frac{-x - xe^{-x}}{2e^{-x} - 2} \cdot \frac{e^x}{e^x} - 1 \\ y(-x) &= \frac{-xe^x - x}{2 - 2e^x} - 1 \\ y(-x) &= \frac{x + xe^x}{2e^x - 2} - 1 \\ y(-x) &= y(x) \\ B_n &= 0 \quad \forall \text{ odd } n > 1 \end{aligned}$$

3 Even integer values $\zeta(2n)$

While the odd integer values of $\zeta(n)$ do not have a closed form expression, the even values do.

Theorem 3.1. We can express all even integer values of ζ using Bernoulli numbers

$$\zeta(2n) = (-1)^{n+1} \frac{(2\pi)^{2n} B_{2n}}{2 \cdot (2n)!} \quad (7)$$

Proof

Starting from the series representation ³ of cotangent

$$\pi x \cot(\pi x) = 1 - 2 \sum_{k=1}^{\infty} \zeta(2k) x^{2k}$$

Using the complex exponential representation ⁴ of cotangent

$$i\pi x + \frac{2i\pi x}{e^{2i\pi x} - 1} = 1 - 2 \sum_{k=1}^{\infty} \zeta(2k) x^{2k}$$

Using the Bernoulli numbers definition ??

$$i\pi x + \sum_{n=0}^{\infty} \frac{B_n}{n!} (2i\pi x)^n = 1 - 2 \sum_{k=1}^{\infty} \zeta(2k) x^{2k}$$

Rearranging and using the first two values of the Bernoulli numbers

$$\begin{aligned} i\pi x + B_0 + B_1(2i\pi x) + \sum_{n=2}^{\infty} \frac{B_n}{n!} (2i\pi x)^n &= 1 - 2 \sum_{k=1}^{\infty} \zeta(2k) x^{2k} \\ i\pi x + 1 + -i\pi x + \sum_{n=2}^{\infty} \frac{B_n}{n!} (2i\pi x)^n &= 1 - 2 \sum_{k=1}^{\infty} \zeta(2k) x^{2k} \\ \sum_{n=2}^{\infty} \frac{B_n}{n!} (2i\pi)^n x^n &= -2 \sum_{k=1}^{\infty} \zeta(2k) x^{2k} \end{aligned}$$

Using the fact the the odd Bernoulli numbers after B_1 are 0

$$\sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} (-1)^k (2\pi)^{2k} x^{2k} = \sum_{k=1}^{\infty} -2\zeta(2k) x^{2k}$$

Comparing coefficients

$$\zeta(2k) = \frac{B_{2k}}{2 \cdot (2k)!} (-1)^{k+1} (2\pi)^{2k} \quad \forall k \in \mathbb{N}_{>0}$$

³See Euler's Sine Product Formula document

⁴ $\cot(x) = i + \frac{2i}{e^{2ix} - 1}$. See Basic trig functions document

3.1 The first few even integer values of ζ

$$\begin{aligned}\zeta(2) &= \frac{\pi^2}{6}, \zeta(4) = \frac{\pi^4}{90}, \zeta(6) = \frac{\pi^6}{945}, \zeta(8) = \frac{\pi^8}{9450}, \zeta(10) = \frac{\pi^{10}}{93555}, \\ \zeta(12) &= \frac{691}{638512875}\pi^{12}, \zeta(14) = \frac{2}{18243225}\pi^{14}, \zeta(16) = \frac{3617}{325641566250}\pi^{16}, \\ \zeta(18) &= \frac{43867}{38979295480125}\pi^{18}, \zeta(20) = \frac{174611}{1531329465290625}\pi^{20}, \zeta(22) = \frac{155366}{13447856940643125}\pi^{22}, \\ \zeta(24) &= \frac{236364091}{201919571963756521875}\pi^{24}, \dots\end{aligned}$$

4 Related functions

4.1 Dirichlet Eta Function

Definition 4.1. *The Dirichlet Eta function is defined as*

$$\eta(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^z} = 1 - \frac{1}{2^z} + \frac{1}{3^z} - \frac{1}{4^z} + \dots$$

Theorem 4.1.

$$\boxed{\eta(z) = (1 - 2^{1-s})\zeta(z)} \quad (8)$$

Proof

Starting from the definition of $\eta(z)$

$$\eta(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^z} = \sum_{k=1}^{\infty} \frac{1}{k^z} - 2 \sum_{k=1}^{\infty} \frac{1}{(2k)^z}$$

Using the definition of $\zeta(z)$

$$\eta(z) = \zeta(z) - \frac{2}{2^z} \zeta(z)$$

$$\eta(z) = (1 - 2^{1-z})\zeta(z)$$

4.1.1 Values

$$\eta(1) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \ln(2) \quad ^5 \quad (9)$$

$$\eta(2) = (1 - 2^{1-2}) \cdot \zeta(2) = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12} \quad (10)$$

⁵This can be seen by taking the Taylor-Expansion of $\ln(x+1)$ and evaluating at $x=1$

4.2 Dirichlet Lambda Function

Definition 4.2. *The Dirichlet Lambda function is defined as*

$$\lambda(z) = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^z} = 1 + \frac{1}{3^z} + \frac{1}{5^z} + \frac{1}{7^z} + \dots$$

Theorem 4.2.

$$\boxed{\lambda(z) = (1 - 2^{-s})\zeta(z)} \quad (11)$$

Proof

Starting from the definition of $\lambda(z)$

$$\lambda(z) = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^z} = \sum_{k=1}^{\infty} \frac{1}{k^z} - \sum_{k=1}^{\infty} \frac{1}{(2k)^z}$$

Using the definition of $\zeta(z)$

$$\lambda(z) = \zeta(z) - \frac{1}{2^z}\zeta(z)$$

$$\lambda(z) = (1 - 2^{-z})\zeta(z)$$

4.2.1 Values

$$\lambda(2) = (1 - 2^{-2}) \cdot \zeta(2) = \frac{3}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{8} \quad (12)$$

5 Integral Forms

5.1 Gamma with Zeta functions

Theorem 5.1. *See the Gamma function document for the definition and properties of the Gamma function.*

$$\boxed{\Gamma(z)\zeta(z) = \int_0^\infty \frac{u^{z-1}}{e^u - 1} du} \quad (13)$$

Proof

Using $\Gamma(z) = n^z \int_0^\infty e^{-nu} u^{z-1} du$ from the Gamma Function document and rearranging

$$\Gamma(z) \frac{1}{n^z} = \int_0^\infty e^{-nu} u^{z-1} du$$

Let n be all integers and take the sum of all equalities

$$\sum_{n=1}^\infty \Gamma(z) \frac{1}{n^z} = \Gamma(z)\zeta(z) = \sum_{n=1}^\infty \int_0^\infty e^{-nu} u^{z-1} du$$

Bringing the summation into the integral since this integral converges uniformly

$$\Gamma(z)\zeta(z) = \int_0^\infty \left(\sum_{n=1}^\infty e^{-nu} \right) u^{z-1} du$$

Notice that $\sum_{n=1}^\infty e^{-nu}$ is a geometric series if $u > 0$

$$\Gamma(z)\zeta(z) = \int_0^\infty \left(\frac{e^{-u}}{1 - e^{-u}} \right) u^{z-1} du$$

Simplifying by multiplying top and bottom by e^u

$$\Gamma(z)\zeta(z) = \int_0^\infty \frac{u^{z-1}}{e^u - 1} du$$

6 Jacobi Theta Function

6.1 Definition

Definition 6.1. *The Jacobi Theta function is defined as*

$$\Theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x}$$

Theorem 6.1. *The Jacobi Theta function has the following functional form*

$$\Theta(x) = \frac{1}{\sqrt{x}} \Theta\left(\frac{1}{x}\right) \quad (14)$$

Proof

Using the Poisson Summation Formula ⁶

$$\Theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x} = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi y^2 x} e^{-2\pi i k y} dy = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi y^2 x - 2\pi i k y} dy$$

Re-writting the exponent

$$-\pi y^2 x - 2\pi i k y = -\pi x \left(y^2 - 2i \frac{k}{x} y \right) = -\pi x \left(y + i \frac{k}{x} \right)^2 - \pi \frac{k^2}{x}$$

Plugging in the factored exponent, we get

$$\Theta(x) = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi x \left(y + i \frac{k}{x} \right)^2 - \pi \frac{k^2}{x}} dy = \sum_{k=-\infty}^{\infty} e^{-\pi \frac{k^2}{x}} \int_{-\infty}^{\infty} e^{-\pi x \left(y + i \frac{k}{x} \right)^2} dy$$

After a change of variables $z = y + i \frac{k}{x}$ we can use the Guassian integral identity as proved before, refer to the Guassian integral document, $\int_{-\infty}^{\infty} e^{-az^2} dz = \sqrt{\frac{\pi}{a}}$, we get:

$$\Theta(x) = \sum_{k=-\infty}^{\infty} e^{-\pi \frac{k^2}{x}} \sqrt{\frac{\pi}{x\pi}} = \frac{1}{\sqrt{x}} \Theta\left(\frac{1}{x}\right)$$

7 Functional Equations

7.1 First Symmetric Form

Theorem 7.1 (Riemann's Symmetric Functional Equation). *The completed zeta function $\xi(s)$ satisfies the symmetric functional equation:*

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \pi^{-\frac{1-s}{2}} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s) \quad (15)$$

⁶ $\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \hat{f}(n)$ is the Poisson summation formula with Fourier transform $\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i k x} dx$. See Fourier Series and Integrals notes.

Proof. Recall the integral definition of the Gamma function:

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty t^{\frac{s}{2}-1} e^{-t} dt$$

Making the substitution $t = \pi n^2 x$ (so $dt = \pi n^2 dx$) for $n \geq 1$:

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty (\pi n^2 x)^{\frac{s}{2}-1} e^{-\pi n^2 x} \pi n^2 dx = \pi^{\frac{s}{2}} n^s \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x} dx$$

Rearranging yields:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} = \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x} dx$$

Summing both sides over all positive integers $n = 1, 2, 3, \dots$ for $\text{Re}(s) > 1$:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \sum_{n=1}^\infty n^{-s} = \sum_{n=1}^\infty \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x} dx$$

By Fubini-Tonelli theorem, we can interchange the sum and integral:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \int_0^\infty x^{\frac{s}{2}-1} \omega(x) dx$$

where $\omega(x) := \sum_{n=1}^\infty e^{-\pi n^2 x} = \frac{\Theta(x)-1}{2}$.

From Jacobi's theta transformation $\Theta(x) = \frac{1}{\sqrt{x}} \Theta\left(\frac{1}{x}\right)$, the function $\omega(x)$ satisfies:

$$\omega(x) = \frac{1}{\sqrt{x}} \omega\left(\frac{1}{x}\right) + \frac{1}{2\sqrt{x}} - \frac{1}{2}$$

Now split the integral into two regions $[0, 1]$ and $[1, \infty)$:

$$\int_0^\infty x^{\frac{s}{2}-1} \omega(x) dx = \int_0^1 x^{\frac{s}{2}-1} \omega(x) dx + \int_1^\infty x^{\frac{s}{2}-1} \omega(x) dx$$

Substitute the modular transformation of $\omega(x)$ into the integral over $[0, 1]$:

$$\begin{aligned} \int_0^1 x^{\frac{s}{2}-1} \omega(x) dx &= \int_0^1 x^{\frac{s}{2}-1} \left(\frac{1}{\sqrt{x}} \omega\left(\frac{1}{x}\right) + \frac{1}{2x^{1/2}} - \frac{1}{2} \right) dx \\ &= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{2} \left[\frac{x^{s/2-1/2}}{s/2-1/2} \right]_0^1 - \frac{1}{2} \left[\frac{x^{s/2}}{s/2} \right]_0^1 \\ &= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{s-1} - \frac{1}{s} \\ &= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{s(s-1)} \end{aligned}$$

In the integral $\int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx$, substitute $u = 1/x$ ($dx = -u^{-2} du$):

$$\int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx = \int_\infty^1 u^{-(\frac{s-1}{2}-1)} \omega(u) \left(-\frac{du}{u^2} \right) = \int_1^\infty u^{\frac{1-s}{2}-1} \omega(u) du$$

Combining both parts, we obtain the fundamental representation:

$$\xi(s) := \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \left(x^{\frac{s}{2}-1} + x^{\frac{1-s}{2}-1}\right) \omega(x) dx \quad (16)$$

Because $\omega(x) \sim e^{-\pi x}$ as $x \rightarrow \infty$, the integral converges absolutely and uniformly for all $s \in \mathbb{C}$, providing the analytic continuation of $\zeta(s)$ to a meromorphic function on $\mathbb{C}$ with a single simple pole at $s = 1$.

Moreover, replacing s with $1-s$ leaves the right-hand side completely invariant:

$$\xi(1-s) = \frac{1}{(1-s)(-s)} + \int_1^\infty \left(x^{\frac{1-s}{2}-1} + x^{\frac{s}{2}-1}\right) \omega(x) dx = \xi(s)$$

which proves the theorem. $\square$

7.2 Second Asymmetric Form

Theorem 7.2 (Reflection Form of the Functional Equation). *For all $s \in \mathbb{C} \setminus \{0, 1\}$:*

$$\boxed{\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)} \quad (17)$$

Proof. From the first functional equation:

$$\zeta(s) = \pi^{s-\frac{1}{2}} \frac{\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)} \zeta(1-s)$$

We simplify the ratio $\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}$ using two classical identities of the Gamma function:

1. Legendre's Duplication Formula:

$$\Gamma\left(\frac{s}{2}\right) \Gamma\left(\frac{s+1}{2}\right) = 2^{1-s} \sqrt{\pi} \Gamma(s) \implies \frac{1}{\Gamma(s/2)} = \frac{2^{s-1}}{\sqrt{\pi}} \frac{\Gamma\left(\frac{s+1}{2}\right)}{\Gamma(s)}$$

2. Euler's Reflection Formula:

$$\Gamma\left(\frac{1-s}{2}\right) \Gamma\left(\frac{1+s}{2}\right) = \frac{\pi}{\sin\left(\pi \frac{1-s}{2}\right)} = \frac{\pi}{\cos\left(\frac{\pi s}{2}\right)}$$

and for $\Gamma(s)$:

$$\Gamma(s) \Gamma(1-s) = \frac{\pi}{\sin(\pi s)} \implies \frac{1}{\Gamma(s)} = \frac{\sin(\pi s)}{\pi} \Gamma(1-s) = \frac{2 \sin\left(\frac{\pi s}{2}\right) \cos\left(\frac{\pi s}{2}\right)}{\pi} \Gamma(1-s)$$

Combining these gives:

$$\begin{aligned} \frac{\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)} &= \Gamma\left(\frac{1-s}{2}\right) \cdot \frac{2^{s-1}}{\sqrt{\pi}} \frac{\Gamma\left(\frac{s+1}{2}\right)}{\Gamma(s)} \\ &= \frac{2^{s-1}}{\sqrt{\pi}} \left[\Gamma\left(\frac{1-s}{2}\right) \Gamma\left(\frac{1+s}{2}\right) \right] \cdot \frac{1}{\Gamma(s)} \\ &= \frac{2^{s-1}}{\sqrt{\pi}} \left[\frac{\pi}{\cos\left(\frac{\pi s}{2}\right)} \right] \cdot \left[\frac{2 \sin\left(\frac{\pi s}{2}\right) \cos\left(\frac{\pi s}{2}\right)}{\pi} \Gamma(1-s) \right] \\ &= 2^s \pi^{-\frac{1}{2}} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \end{aligned}$$

Multiplying by $\pi^{s-1/2}\zeta(1-s)$ yields:

$$\zeta(s) = \pi^{s-\frac{1}{2}} \cdot 2^s \pi^{-\frac{1}{2}} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)$$

□

8 Zeros of the Riemann Zeta Function and the Riemann Hypothesis

8.1 Trivial Zeros

Using the reflection functional equation:

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)$$

Consider negative even integers $s = -2m$ for $m \in \{1, 2, 3, \dots\}$:

1. The factor $\sin\left(\frac{\pi(-2m)}{2}\right) = \sin(-\pi m) = 0$.
2. The factor $\Gamma(1 - (-2m)) = \Gamma(1 + 2m) = (2m)!$ is a positive real number (finite and non-zero).
3. The factor $\zeta(1 - (-2m)) = \zeta(2m + 1)$ is evaluated in the region of absolute convergence $\text{Re}(s) > 1$, where the Euler product guarantees $\zeta(2m + 1) > 1 \neq 0$.

Therefore:

$$\boxed{\zeta(-2m) = 0 \quad \text{for all } m \in \{1, 2, 3, \dots\}} \quad (18)$$

These are called the **trivial zeros** of the Riemann zeta function.

Remark 8.1 (Negative odd integers). For negative odd integers $s = -(2m - 1)$, $\sin\left(\frac{\pi s}{2}\right) = \sin\left(-\frac{(2m-1)\pi}{2}\right) = (-1)^m \neq 0$, and using the connection to Bernoulli numbers from Theorem ??:

$$\zeta(-(2m - 1)) = -\frac{B_{2m}}{2m} \neq 0$$

In particular, $\zeta(-1) = -\frac{1}{12}$, $\zeta(-3) = \frac{1}{120}$, and $\zeta(-5) = -\frac{1}{252}$.

8.2 The Critical Strip and Non-Trivial Zeros

1. **Region** $\text{Re}(s) > 1$: The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ converges absolutely and non-zero, so $\zeta(s)$ has no zeros for $\text{Re}(s) > 1$.
2. **Region** $\text{Re}(s) < 0$: By the functional equation, the only zeros for $\text{Re}(s) < 0$ are the trivial zeros at $s = -2, -4, -6, \dots$.
3. **The Critical Strip**: Consequently, all other zeros of $\zeta(s)$ must lie strictly in the strip:

$$\mathcal{S} = \{s \in \mathbb{C} : 0 \leq \text{Re}(s) \leq 1\}$$

These are called the **non-trivial zeros**.

4. **Zero-Free Boundary** $\text{Re}(s) = 1$: In 1896, Jacques Hadamard and Charles Jean de la Vallée Poussin independently proved that:

$$\zeta(1 + it) \neq 0 \quad \text{for all } t \in \mathbb{R}$$

By the functional symmetry $s \leftrightarrow 1 - s$, $\zeta(0 + it) \neq 0$ as well. This zero-free boundary is mathematically equivalent to the **Prime Number Theorem**:

$$\pi(x) \sim \frac{x}{\ln x} \quad \text{as } x \rightarrow \infty$$

8.3 Statement of the Riemann Hypothesis

The Riemann Hypothesis (Bernhard Riemann, 1859)

All non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the **critical line**:

$$\boxed{\text{Re}(s) = \frac{1}{2}} \quad (19)$$

That is, if $\zeta(s) = 0$ with $0 < \text{Re}(s) < 1$, then $s = \frac{1}{2} + i\gamma$ for some $\gamma \in \mathbb{R}$.

The first few non-trivial zeros on the critical line have imaginary parts:

$$\gamma_1 \approx 14.134725 \dots$$

$$\gamma_2 \approx 21.022040 \dots$$

$$\gamma_3 \approx 25.010857 \dots$$

$$\gamma_4 \approx 30.424876 \dots$$

More than 10^{13} zeros have been verified numerically to lie on the critical line, but a general proof remains one of the most famous open problems in mathematics (a Clay Millennium Prize problem).